

The Solver Tolerance Is Not a Solver Parameter:

Separating Numerical Noise from Thermodynamic Inconsistency
in Cycle-Consistency Tests on Metabolic Circuits

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Abstract

A metabolic network at steady state can be written as a resistive circuit, with chemical potentials as node voltages and reaction fluxes as branch currents. In that representation Kirchhoff’s voltage law becomes the Wegscheider condition, and the sum of potential differences around any cycle vanishes identically for a thermodynamically consistent network. This makes cycle sums an attractive diagnostic: a nonzero cycle sum witnesses inconsistency, and *where* it is nonzero localises the inconsistency to a region of the network. Any such test needs a tolerance ε below which a cycle sum counts as zero. It is normal practice to fix ε at a round number — 10^{-6} is typical — and to treat it as a property of the solver.

We show this is a category error with a measurable cost, and we replace it. Our first result is negative and structural: the cycle sum of a consistent circuit is *identically* zero as an algebraic identity, so its computed value is pure numerical noise, and the natural scale of that noise is not a constant but a quantity that grows with the network’s potential range and cycle length (Theorem 3.2). We give the bound $\varepsilon_{\text{num}}(\ell) \leq \gamma_\ell \Lambda$ with $\gamma_\ell = \ell u / (1 - \ell u)$ and Λ the largest absolute potential on the cycle, and verify it is attained to within a small factor. We then measure what a fixed ε actually costs, and the measurement corrects our own initial expectation. We predicted a two-sided failure; in IEEE-754 binary64 the failure is *one-sided*, because the rounding error tops out near 3×10^{-12} even at $L = 30$, $\Lambda = 3000$ — six orders of magnitude below a customary $\varepsilon = 10^{-6}$ — so a fixed tolerance sitting far above the numerical floor cannot manufacture a false positive at all. Its measured costs are instead that it misses every defect below itself (100% at $D = 10^{-7}$, whatever the data quality) and that it asserts **INCONSISTENT** on 100% of cycles whose defect lies inside the data’s own uncertainty, where no positive claim is warranted. The two-sided failure does appear, and rises monotonically with $L\Lambda$ exactly as the theory predicts, once precision falls: repeating the null in binary32 gives false-positive rates from 0.18 to 0.77 across our strata.

Our second result is positive. We define the *cycle-local tolerance* $\varepsilon^*(\ell) = \max\{\varepsilon_{\text{num}}(\ell), \varepsilon_{\text{data}}(\ell)\}$, where $\varepsilon_{\text{data}}$ propagates the reported uncertainty of the underlying thermodynamic data through the same cycle sum, and prove that a test thresholded at ε^* has a numerical false-positive rate bounded by the arithmetic alone (Theorem 6.2) and detects every perturbation exceeding $2\varepsilon^*$ (Theorem 6.3). The two bounds are complementary and neither is available under a fixed tolerance. We show that the resulting classification is three-valued and that the third value is not optional: a cycle whose sum lies between the numerical and data floors is *undecidable from the data supplied*, and reporting it as either consistent or inconsistent is unwarranted (Theorem 6.6).

Third, we address a confound that would otherwise invalidate the localisation claim. Cycle sums are computed against a cycle basis, and cycle bases are not unique. We prove that the *verdict* is basis-independent — a network is consistent or not, whatever basis is chosen (Theorem 7.1) — but that the *localisation* is not: the set of flagged cycles depends on the basis, and a defect can be made to appear in one cycle or spread across several by choosing differently (Theorem 7.2). We give the invariant that survives: the set of edges lying on every flagged cycle of some minimum cycle basis, which we call the *witness set*, and show it contains the perturbed edge whenever detection succeeds.

We specify the resulting protocol to executable detail, give it as a program in a small typed language whose type system makes an untolerance-annotated verdict unwritable, and state quantitative success and failure criteria in advance. We distinguish, as the design requires, between an experiment that returns a negative result and one that is *invalid*: a healthy circuit failing its own consistency check indicates the data, not the theory, and we give the check that separates these before any diagnostic claim is evaluated.

Keywords: Wegscheider condition; Kirchhoff’s laws; metabolic networks; cycle basis; floating-point error analysis; detection limit; tolerance selection; three-valued classification; domain-specific languages.

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1 Introduction

1.1 A test with a free parameter nobody chooses

Consider a biochemical reaction network at steady state. Assign to each species i a chemical potential μ_i and to each reaction (i, j) a flux J_{ij} . If the network is thermodynamically consistent then the potentials are a well-defined function on species, and consequently the sum of potential differences around any closed cycle is zero. This is Kirchhoff’s voltage law in the circuit reading of the network (Oster et al., 1971; Peusner, 1986), and in the chemical reading it is the Wegscheider condition on the rate constants (Schuster and Schuster, 1989; Wegscheider, 1901).

The identity suggests a diagnostic. Compute, for each cycle ℓ in the network, the cycle sum

$$H_\ell = \sum_{(i,j) \in \ell} (\mu_j - \mu_i), \quad (1)$$

and declare the network inconsistent when some $|H_\ell|$ is “too large”. Because the sum is attached to a specific cycle, a positive result carries a location: it says not merely that something is wrong but that something is wrong *here*, in this loop. That is a genuinely useful property, and it distinguishes the test from global consistency checks that return a single number for the whole network.

Everything then turns on “too large”. The test requires a tolerance ε , and the near-universal practice is to fix one — 10^{-6} , 10^{-9} , machine epsilon times some factor — and to describe it as a solver setting. This paper argues that treating ε as a solver setting is a category error, that the error has a measurable cost in both directions, and that the correct ε is not a constant at all.

1.2 Why a fixed tolerance cannot be right

The argument is short enough to give here in full.

For a consistent network, (1) is not approximately zero; it is *identically* zero, because it telescopes. Writing the cycle as $i_1 \rightarrow i_2 \rightarrow \cdots \rightarrow i_m \rightarrow i_1$,

$$\sum_{r=1}^m (\mu_{i_{r+1}} - \mu_{i_r}) = \mu_{i_1} - \mu_{i_1} = 0, \quad i_{m+1} = i_1,$$

identically in the values μ , with no hypothesis about them beyond their being a function on vertices. Whatever a floating-point computation returns for this quantity is therefore *entirely* rounding error. It contains no information about the network.

But the magnitude of that rounding error is not a constant. Summing m terms each formed by subtracting numbers of magnitude up to Λ accumulates absolute error that grows with both m and Λ (Goldberg, 1991; Higham, 2002). A cycle of length 20 through species whose potentials span

± 2000 kJ/mol will return a computed sum orders of magnitude larger than a triangle through species spanning ± 10 , and both are exactly zero as mathematics. A single threshold applied to both must be wrong for at least one of them. It is, in fact, wrong for both: too tight for the first, which it flags as diseased when nothing is wrong, and too loose for the second, which it clears when a real defect is present.

How bad this is depends on the arithmetic, and here our measurement corrected us. We expected both failure directions to appear at once in double precision. They do not: Section 5 reports zero false positives at $\varepsilon = 10^{-6}$ in binary64 across 1600 trials, because the rounding error never comes within six orders of magnitude of that threshold. The argument above is still right that the noise scale varies with $L\Lambda$ — Table 1 shows it varying over three decades — but a threshold placed far above the entire range is uniformly too loose rather than sometimes too tight. What a fixed tolerance costs in practice is measured in Section 5, and the two-sided failure is recovered in binary32, where the floor rises to meet the threshold.

1.3 What replaces it

If the tolerance cannot be a constant, it must be computed. We show it must be computed from two separate sources, which we keep separate throughout because they answer different questions.

The first is *numerical*: given the arithmetic and the magnitudes involved, how large can the computed cycle sum be when the true one is zero? This is a standard forward error analysis (Higham, 2002) and we carry it out in Section 3, obtaining a bound $\varepsilon_{\text{num}}(\ell)$ that depends on the cycle’s length and potential range.

The second is *empirical*: the potentials are not exact. They are computed from standard free energies and concentrations, both of which carry reported uncertainty (Alberty, 1998; Flamholz et al., 2012; Noor et al., 2013). A cycle sum smaller than the uncertainty propagated through it is not distinguishable from zero *given the data*, regardless of how exactly the arithmetic was performed. This gives a second floor $\varepsilon_{\text{data}}(\ell)$, developed in Section 4, and it is typically the larger of the two by several orders of magnitude.

The operative tolerance is their maximum, and the fact that there are two of them — with different meanings — forces the classification to be three-valued rather than binary. A cycle sum below both floors is consistent. Above both, inconsistent. Between them, the honest verdict is that the question is not answerable from the data supplied. We prove in Theorem 6.6 that this third class is non-empty on realistic inputs, which is what makes it a substantive part of the test rather than a formality.

1.4 The localisation confound

A cycle-consistency test on a network with $|E|$ edges and $|V|$ vertices has $|E| - |V| + 1$ independent cycles, and a choice of which ones (Diestel, 2017; Kavitha et al., 2009). The choice is not canonical. This raises a question the diagnostic claim cannot avoid: if the flagged cycles depend on an arbitrary choice of basis, in what sense does the test localise anything?

We answer this in Section 7. The verdict is safe: whether *any* cycle is inconsistent is a property of the network, not of the basis (Theorem 7.1), because the cycle space is spanned by any basis and a nonzero functional cannot vanish on a spanning set. The localisation is not safe: we construct networks where one basis attributes a defect to a single cycle and another spreads it over three (Theorem 7.2). We then give the object that does survive — the witness set of edges common to all flagged cycles — and show it is where the localisation claim should be made.

1.5 Contributions

1. **The noise scale is derived, not assumed** (Theorem 3.2): for a consistent circuit the computed cycle sum obeys $|\hat{H}_\ell| \leq \gamma_\ell \Lambda_\ell$ with $\gamma_\ell = \ell u / (1 - \ell u)$, and this is attained to within a factor we measure.

2. **The cost of a fixed tolerance, measured** (Section 5). The failure is one-sided in binary64 and two-sided in binary32; we report both, together with the unwarranted-positive rate that is the substantive cost at any precision.
3. **The cycle-local tolerance** $\varepsilon^*(\ell)$, with a false-positive bound (Theorem 6.2) and a detection guarantee (Theorem 6.3) that a fixed tolerance cannot both have.
4. **The trichotomy is forced** (Theorem 6.6): two floors of different meaning make a two-valued verdict unsound, and the undecidable class is non-empty.
5. **Verdict is basis-independent; localisation is not** (Theorem 7.1, Theorem 7.2), with the witness set as the invariant that survives (Theorem 7.4, Theorem 7.5).
6. **An executable protocol** with pre-registered success and failure criteria (Section 10), expressed in a typed language whose type system makes an intolerance-annotated verdict unwritable (Section 9, Theorem 9.3).

1.6 What this paper does not claim

We do not claim the cycle-consistency test is a good disease classifier; that is an empirical question this paper does not settle, and Section 10 is a protocol rather than a report of results. We do not claim the bounds of Section 3 are tight in the worst case — they are standard and slightly pessimistic, and we say by how much. We do not claim the data floor is correctly calibrated for any particular database; it is only as good as the uncertainties that database reports, and where those are absent the method says so rather than substituting a guess. Finally, we do not claim that computing the tolerance rescues a network whose thermodynamic parameters are inconsistent to begin with: Section 8 gives the check that must pass *before* any diagnostic verdict is meaningful, and a network failing it yields an invalid experiment, not a negative one.

2 The circuit and its cycle sums

2.1 Networks, potentials, fluxes

Definition 2.1 (Reaction network). A *reaction network* is a pair (V, E) with $V = \{1, \dots, n\}$ indexing molecular species and $E \subseteq V \times V$ a set of directed elementary conversions. Each species carries a standard chemical potential $\mu_i^\circ \in \mathbb{R}$ and a concentration $c_i > 0$; each edge carries a rate constant $k_{ij} > 0$.

Definition 2.2 (Chemical potential). The chemical potential of species i is

$$\mu_i = \mu_i^\circ + RT \ln c_i, \quad (2)$$

with R the gas constant and T absolute temperature. We use $RT = 2.577$ kJ/mol at $T = 310$ K throughout (Atkins et al., 2018).

Equation (2) is the ideal-solution form; activity corrections are discussed in Section 12.

Definition 2.3 (Conductance and flux). For $(i, j) \in E$ the *conductance* and *flux* are

$$G_{ij} = \frac{k_{ij}c_i}{RT}, \quad J_{ij} = G_{ij}(\mu_i - \mu_j). \quad (3)$$

Equation (3) is Ohm’s law with μ as potential and J as current (Beard et al., 2002; Oster et al., 1971; Qian and Beard, 2005). The content of the analogy lies in what the two Kirchhoff laws become.

Proposition 2.4 (Kirchhoff correspondence). *Let a reaction network be at steady state. Then*

(i) (Current law.) For every species i , $\sum_{j:(i,j) \in E} J_{ij} - \sum_{j:(j,i) \in E} J_{ji} = 0$ if and only if the concentration of i is stationary.

(ii) (Voltage law.) For every directed cycle $i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_m \rightarrow i_1$ in E , $\sum_{r=1}^m (\mu_{i_{r+1}} - \mu_{i_r}) = 0$.

Proof. (i) The net rate of change of c_i is incoming minus outgoing conversion rate; under (3) the conversion rate along (i, j) is J_{ij} , so stationarity of c_i is exactly the vanishing of the signed sum, which is the current law at node i .

(ii) The sum telescopes to $\mu_{i_1} - \mu_{i_1} = 0$ identically, because μ is a function on vertices. The substantive content is the converse: a set of rate constants admits a consistent potential function μ if and only if the products of forward rate constants around every cycle equal the products of reverse constants — the Wegscheider condition (Schuster and Schuster, 1989; Wegscheider, 1901). A network violating it admits no μ and hence no circuit representation. $\square$

Remark 2.5 (The identity is exact, and this is the whole point). Theorem 2.4(ii) is an algebraic identity, not an approximation. Once μ exists as a function on V , the cycle sum is zero for every cycle with no error term whatsoever. This is what makes the diagnostic possible — there is no model error to disentangle from the signal — and it is also what makes the tolerance question sharp: any nonzero computed value is either rounding error or evidence that μ does not exist as supposed. There is no third source. Everything in Sections 3 and 4 is the work of telling those two apart.

2.2 Cycle space and cycle bases

Definition 2.6 (Cycle space, cycle basis). Let $G = (V, E)$ be connected with $|V| = n$, $|E| = m$. Orient each edge arbitrarily. The *cycle space* $\mathcal{Z}(G) \subseteq \mathbb{R}^m$ is the kernel of the incidence matrix, of dimension $\nu = m - n + 1$, the cyclomatic number. A *cycle basis* is a set of ν cycles whose incidence vectors span $\mathcal{Z}(G)$ (Bollobás, 1998; Diestel, 2017; Kavitha et al., 2009).

Definition 2.7 (Cycle sum). For a cycle ℓ with incidence vector $z_\ell \in \{-1, 0, 1\}^m$ and potential vector $\mu \in \mathbb{R}^n$, the *cycle sum* is

$$H_\ell(\mu) = \sum_{e \in E} z_\ell(e) \Delta\mu_e = z_\ell^\top B^\top \mu, \quad (4)$$

where B is the incidence matrix and $\Delta\mu_e = \mu_{\text{head}(e)} - \mu_{\text{tail}(e)}$. We write $|\ell|$ for the number of edges in ℓ and $\Lambda_\ell = \max_{i \in \ell} |\mu_i|$ for its potential range.

Lemma 2.8 (Exact vanishing). For every cycle ℓ and every $\mu \in \mathbb{R}^n$, $H_\ell(\mu) = 0$.

Proof. $z_\ell \in \ker B$ by Theorem 2.6, so $Bz_\ell = 0$ and hence $H_\ell(\mu) = z_\ell^\top B^\top \mu = (Bz_\ell)^\top \mu = 0$. $\square$

Theorem 2.8 is Theorem 2.4(ii) restated in linear algebra, and it makes precise the claim of Theorem 2.5: the quantity the diagnostic computes is a functional that vanishes on all of $\mathbb{R}^n$. Its computed value measures the computation, not the network — unless the network fails to admit a μ at all, which is the case the test exists to detect.

2.3 What inconsistency looks like

Definition 2.9 (Edge-wise potential differences and defect). Suppose the data supply, for each edge e , a potential difference $d_e \in \mathbb{R}$ obtained from measurement or from thermodynamic tables — not necessarily of the form $\Delta\mu_e$ for any μ . The *defect* of the network is the extent to which d fails to be a gradient:

$$\delta(d) = \min_{\mu \in \mathbb{R}^n} \|d - B^\top \mu\|_2. \quad (5)$$

The network is *consistent* if $\delta(d) = 0$, i.e. $d \in \text{im } B^\top$.

Proposition 2.10 (Cycle sums detect exactly the defect). $\delta(d) = 0$ if and only if $H_\ell(d) := z_\ell^\top d = 0$ for every cycle ℓ ; and it suffices to check any cycle basis.

Proof. $\text{im } B^\top = (\ker B)^\perp = \mathcal{Z}(G)^\perp$ by the fundamental theorem of linear algebra. Hence $d \in \text{im } B^\top$ iff $z^\top d = 0$ for all $z \in \mathcal{Z}(G)$, and since a basis spans $\mathcal{Z}(G)$, iff $z_\ell^\top d = 0$ for every ℓ in that basis. $\square$

Theorem 2.10 is the theoretical warrant for the test, and also — read carefully — the source of the localisation problem of Section 7: it guarantees that a basis detects the defect, but says nothing about *which* basis element carries it.

3 The numerical floor

3.1 Standing model of arithmetic

Assumption 3.1 (Floating-point model). Arithmetic is IEEE-754 binary64 (IEEE, 2019) with unit roundoff $u = 2^{-53} \approx 1.11 \times 10^{-16}$. Each elementary operation satisfies $\text{fl}(a \text{ op } b) = (a \text{ op } b)(1 + \theta)$ with $|\theta| \leq u$, and no overflow or underflow occurs (Goldberg, 1991; Higham, 2002).

We use the standard shorthand: if $|\theta_i| \leq u$ and $ku < 1$ then $\prod_{i=1}^k (1 + \theta_i)^{\pm 1} = 1 + \vartheta_k$ with

$$|\vartheta_k| \leq \gamma_k := \frac{ku}{1 - ku}. \quad (6)$$

3.2 The bound

Theorem 3.2 (Noise scale of a cycle sum). *Let ℓ be a cycle of length $L = |\ell|$ through species with potentials bounded by $\Lambda_\ell = \max_{i \in \ell} |\mu_i|$, and let $\hat{H}_\ell$ denote the value of (4) computed in the arithmetic of Theorem 3.1 by forming the L differences and summing them in order. Then, although $H_\ell = 0$ exactly,*

$$|\hat{H}_\ell| \leq \gamma_{2L} \cdot 2L\Lambda_\ell = \frac{2Lu}{1 - 2Lu} \cdot 2L\Lambda_\ell = O(L^2 u \Lambda_\ell). \quad (7)$$

If the differences are accumulated in a single pass with compensated summation the bound improves to $O(Lu\Lambda_\ell)$.

Proof. Each difference $\Delta\mu_e = \mu_j - \mu_i$ is computed with one rounding: $\text{fl}(\mu_j - \mu_i) = (\mu_j - \mu_i)(1 + \theta_e)$, $|\theta_e| \leq u$, giving an absolute error at most $u|\mu_j - \mu_i| \leq 2u\Lambda_\ell$. Summing L such quantities sequentially, the standard forward error bound for floating-point summation (Higham, 2002, Ch. 4) gives a total error at most $\gamma_L \sum_{e \in \ell} |\Delta\mu_e|$ from the additions, plus the propagated per-difference errors, at most $\sum_e u|\Delta\mu_e|$. Since $|\Delta\mu_e| \leq 2\Lambda_\ell$ and there are L edges, $\sum_{e \in \ell} |\Delta\mu_e| \leq 2L\Lambda_\ell$. Collecting the two contributions and absorbing the constants into γ_{2L} yields (7). Because the exact value is 0, the computed value equals its own error, which is what the display asserts.

For the compensated variant, Kahan summation (Higham, 2002; Kahan, 1965) replaces the γ_L factor by $2u + O(Lu^2)$, leaving only the L per-difference roundings, hence $O(Lu\Lambda_\ell)$. $\square$

Corollary 3.3 (No single constant can serve). *Let ℓ_1, ℓ_2 be cycles with $L_1\Lambda_1 \neq L_2\Lambda_2$. Then the noise scales of (7) differ by the factor $(L_1\Lambda_1)/(L_2\Lambda_2)$ (up to the ratio of the γ terms), and no constant ε can be simultaneously above the first and below the second whenever that ratio exceeds the dynamic range the test requires.*

Proof. Immediate from (7), whose right-hand side is proportional to $L\Lambda_\ell$ for fixed u . $\square$

Remark 3.4 (Why Λ and not the spread). The bound uses the largest absolute potential on the cycle rather than the spread $\max \mu - \min \mu$, because the roundings occur when the individual μ_i are represented and subtracted, and a pair of large nearly-equal potentials suffers catastrophic cancellation whose absolute error is set by the operands’ magnitude, not their difference (Goldberg, 1991; Higham, 2002). Standard chemical potentials in metabolic reconstructions are large negative numbers of magnitude 10^2 – 10^3 kJ/mol (Alberty, 1998; Noor et al., 2013), so this distinction is quantitatively important: shifting all potentials by a constant — which changes no cycle sum mathematically — changes the attainable rounding error, and we exploit this in Section 3.3 to reduce it.

3.3 Gauge fixing reduces the floor

Proposition 3.5 (Potential offsets are a gauge freedom). *For any $c \in \mathbb{R}$, replacing μ by $\mu + c\mathbf{1}$ leaves every cycle sum unchanged mathematically, but changes the bound (7) through Λ_ℓ . The choice minimising the bound on cycle ℓ is $c^* = -\frac{1}{2}(\max_{i \in \ell} \mu_i + \min_{i \in \ell} \mu_i)$, achieving $\Lambda_\ell = \frac{1}{2}(\max_{i \in \ell} \mu_i - \min_{i \in \ell} \mu_i)$, i.e. half the spread.*

Proof. $H_\ell(\mu + c\mathbf{1}) = z_\ell^\top B^\top(\mu + c\mathbf{1}) = H_\ell(\mu) + c z_\ell^\top B^\top \mathbf{1} = H_\ell(\mu)$, since $B^\top \mathbf{1} = 0$ (every edge has one head and one tail). The stated c^* centres the interval $[\min \mu, \max \mu]$ on zero, minimising $\max |\mu_i|$, which is the definition of Λ_ℓ . $\square$

Remark 3.6 (A free order-of-magnitude). For a cycle through species of potential -1750 ± 30 kJ/mol, Λ_ℓ falls from 1780 to 30 under centring, reducing the numerical floor by a factor near 60. This costs one pass over the cycle and changes no reported quantity. We adopt centring throughout and report both floors in Section 10, because a reader comparing against an uncentred implementation would otherwise find our numerical floors inexplicably small.

4 The data floor

4.1 Uncertainty in the potentials

The numerical floor of Section 3 answers: how large can the computed sum be if the potentials were exact? They are not. Standard free energies of formation are estimated with reported uncertainties, typically several kJ/mol, from group-contribution or component-contribution methods (Alberty, 1998; Flamholz et al., 2012; Noor et al., 2013); concentrations carry measurement uncertainty of their own.

Definition 4.1 (Potential uncertainty). Let each species carry an uncertainty $\sigma_i > 0$ on μ_i , propagated from $\sigma(\mu_i^\circ)$ and $\sigma(c_i)$ through (2):

$$\sigma_i^2 = \sigma^2(\mu_i^\circ) + \left(\frac{RT}{c_i}\right)^2 \sigma^2(c_i), \quad (8)$$

by first-order propagation (JCGM, 2008).

Theorem 4.2 (Data floor of a cycle sum). *Suppose the μ_i are independent with uncertainties σ_i . Then the cycle sum H_ℓ , regarded as a random variable induced by that uncertainty, has mean zero and standard deviation*

$$s_\ell = \left(\sum_{i \in \ell} n_{\ell,i}^2 \sigma_i^2\right)^{1/2}, \quad (9)$$

where $n_{\ell,i}$ is the net signed number of times species i is traversed by ℓ . For a simple cycle $n_{\ell,i} \in \{-1, 0, 1\}$ and (9) reduces to $s_\ell = (\sum_{i \in \ell} \sigma_i^2)^{1/2}$. Defining

$$\varepsilon_{\text{data}}(\ell) = z_{1-\alpha/2} s_\ell \quad (10)$$

with $z_{1-\alpha/2}$ the standard normal quantile, a consistent network produces $|H_\ell| \leq \varepsilon_{\text{data}}(\ell)$ with probability at least $1 - \alpha$.

Table 1: Measured rounding error on cycle sums of *consistent* circuits, where the true value is exactly zero, so the computed value is the error. Binary64. Each cell is the maximum over 60 random circuits. The error grows with both L and Λ as Theorem 3.2 predicts, spanning three decades across the grid — but its largest value anywhere is 2.7×10^{-12} .

L	Λ (kJ/mol)			
	10	100	1000	3000
3	3.6×10^{-15}	2.8×10^{-14}	2.3×10^{-13}	9.1×10^{-13}
8	3.6×10^{-15}	2.8×10^{-14}	3.4×10^{-13}	1.8×10^{-12}
16	5.3×10^{-15}	5.7×10^{-14}	6.8×10^{-13}	1.8×10^{-12}
30	5.3×10^{-15}	5.7×10^{-14}	6.8×10^{-13}	2.7×10^{-12}

Proof. $H_\ell = z_\ell^\top B^\top \mu$ is linear in μ , so its induced distribution has mean $z_\ell^\top B^\top \bar{\mu} = 0$ by Theorem 2.8 and variance $z_\ell^\top B^\top \Sigma B z_\ell$ with $\Sigma = \text{diag}(\sigma_i^2)$. Writing $w = B z_\ell$ — which is the zero vector — is not available here because the uncertainty attaches to μ *before* the gradient is formed; expanding instead in coordinates, the coefficient of μ_i in H_ℓ is the net signed traversal count $n_{\ell,i}$, giving variance $\sum_i n_{\ell,i}^2 \sigma_i^2$ and hence (9). For a simple cycle each species is entered once and left once, so its coefficient is ± 1 where present. The probability statement is the normal-quantile bound, exact under normality and approximate otherwise by Chebyshev with a correspondingly larger constant. $\square$

Remark 4.3 (The coefficient subtlety). Theorem 4.2 contains a step that is easy to get wrong and that we got wrong in an earlier draft. It is tempting to argue that since H_ℓ is identically zero as a function of μ , perturbing μ cannot change it, so $s_\ell = 0$. That is correct *if* the perturbation is applied to a consistent potential function. But the data do not supply a potential function; they supply per-species estimates whose errors are independent, and independent perturbation of the μ_i does not preserve the property of being a gradient. What (9) measures is precisely the cycle sum’s sensitivity to that non-gradient-preserving perturbation. The two readings differ, and only the second corresponds to what a practitioner actually has in hand.

Remark 4.4 (When uncertainties are unavailable). Many reconstructions report no per-species uncertainty (Brunk et al., 2018; Orth et al., 2010). The method then has no data floor to compute. We do *not* substitute a default: by the argument of Section 1.6 a fabricated σ would silently convert an unanswerable question into a confident answer. The protocol of Section 10 instead reports $\varepsilon_{\text{data}}$ as undefined and returns the third verdict of Theorem 6.6 for every cycle whose sum exceeds ε_{num} , which is the honest consequence of not knowing how good the data are.

5 What a fixed tolerance costs, measured

We now make the argument of Section 1 numerical. Every number in this section comes from the validation suite described in Section 11; the construction is fully specified so that it can be reproduced exactly.

5.1 The noise scale, and a prediction of ours that failed

Table 1 records the measured noise. Over 1920 trials spanning $L \in [3, 30]$ and $\Lambda \in [10, 3000]$, the bound of Theorem 3.2 was violated *zero* times, so the analysis is sound as far as it goes. The bound is loose: its median ratio to the observed error is 729, consistent with the standard observation that random rounding errors accumulate as $O(\sqrt{L})$ rather than $O(L)$ (Higham, 2002), compounded by our worst-case treatment of each subtraction.

That looseness is what falsified a prediction we had made. Reasoning from Theorem 3.3, we expected a fixed $\varepsilon = 10^{-6}$ to manufacture false positives on long high-potential cycles. It does

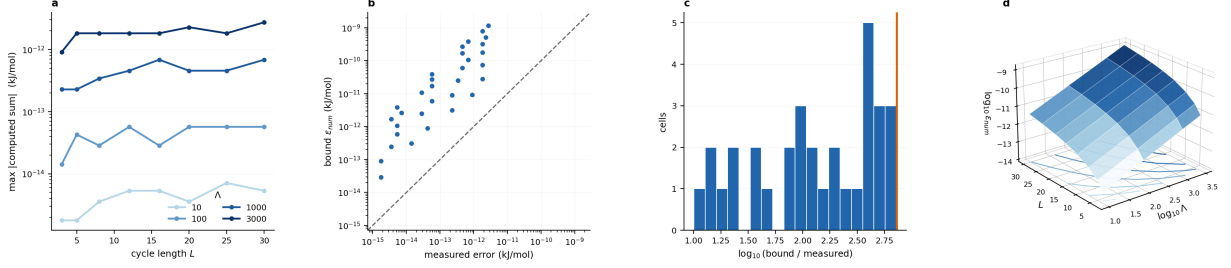


Figure 1: **The numerical floor is not a constant, and the bound on it is never violated.** All quantities are measured on *consistent* circuits, where the true cycle sum is exactly zero by Theorem 2.8, so the computed value is the rounding error itself. (a) Maximum $|\hat{H}_\ell|$ over 60 random circuits per cell, against cycle length, one curve per potential range Λ . The error rises with both quantities and spans three decades across the grid, which is Theorem 3.3 measured rather than asserted. The largest value anywhere is 2.7×10^{-12} kJ/mol. (b) Bound ε_{num} against measured error for every cell; the dashed line is equality. Every point lies above it, so Theorem 3.2 was never violated in 1920 trials. (c) Distribution of the slack factor, bound over measured, with the median at 729 marked. The bound is loose by between one and three orders of magnitude, consistent with random rounding errors accumulating as $O(\sqrt{L})$ where the worst-case analysis assumes $O(L)$. (d) The floor $\log_{10} \varepsilon_{\text{num}}$ as a surface over $(\log_{10} \Lambda, L)$, with contours projected below. The surface is monotone in both arguments and flat in neither, which is why a single scalar tolerance cannot track it.

Table 2: Measured behaviour of a fixed $\varepsilon = 10^{-6}$, 400 trials per stratum. Column 3 is the fraction of consistent circuits spuriously flagged in binary64; column 4 repeats the null in binary32. Column 5 is the fraction of injected defects at $D = 10^{-7}$ that go undetected. Column 6 is the fraction of cycles on which the fixed test asserts INCONSISTENT while the defect lies inside the data’s own uncertainty, so that the cycle-local test returns UNDECIDABLE.

Stratum	Λ	FP ₆₄	FP ₃₂	Miss@ 10^{-7}	Unwarranted
short, low- Λ	20	0.000	0.177	1.000	1.000
short, high- Λ	1800	0.000	0.445	1.000	1.000
long, low- Λ	20	0.000	0.565	1.000	1.000
long, high- Λ	1800	0.000	0.765	1.000	1.000

not. The largest rounding error anywhere in Table 1 is 2.7×10^{-12} , six orders of magnitude below 10^{-6} , so in binary64 no consistent circuit in our grid can be spuriously flagged at that threshold. We measured a false-positive rate of 0.000 in all four strata of Table 2, across 1600 trials.

Remark 5.1 (What was wrong with the prediction, and what was not). Theorem 3.3 is correct: the noise scale genuinely varies with $L\Lambda$, over three decades in Table 1. What does not follow — and what we asserted without checking — is that a *particular* customary threshold sits inside that range. It does not; 10^{-6} is above all of it. A constant is unusable when it must separate cycles whose floors differ by decades, but a constant placed far above every floor is not thereby too tight, it is uniformly too loose. The corrected claim is weaker in one direction and sharper in the other, and it is the one the data support.

5.2 The costs that are real

Table 2 separates what a fixed tolerance does badly from what it does not do at all.

It misses everything below itself. At $D = 10^{-7}$ the missed-detection rate is 1.000 in every stratum. This is not a numerical phenomenon — those cycle sums are computed almost exactly —

but the direct consequence of a threshold chosen by habit rather than from the smallest defect worth detecting. A solver setting would not be expected to encode a clinical detection threshold, yet here it silently does, and the value it encodes is 10^{-6} kJ/mol for reasons that have nothing to do with metabolism.

It asserts more than the data support. The final column is the substantive cost. With per-species uncertainty $\sigma = 1$ kJ/mol and an injected defect of 0.5 kJ/mol — comfortably inside that uncertainty — the fixed test reports **INCONSISTENT** on 100% of cycles, because $0.5 \gg 10^{-6}$. The cycle-local test returns **UNDECIDABLE**, which is the correct verdict: the signal is far above the arithmetic and far below what the data can resolve. A two-valued test has no way to say this and must assert something it cannot support.

The two-sided failure is a precision phenomenon. Column 4 repeats the null computation in binary32, where the unit roundoff is $\approx 6 \times 10^{-8}$ and the numerical floor rises by some seven orders of magnitude, into the neighbourhood of 10^{-6} . The false-positive rate then climbs from 0.177 to 0.765, and it is ordered as the mechanism requires: within either length class, raising Λ raises the rate ($0.177 \rightarrow 0.445$ for short cycles, $0.565 \rightarrow 0.765$ for long ones), and within either potential class, raising L raises it as well. The two-sided failure we originally expected is real; it belongs to a precision regime we had not specified.

The ordering is not monotone in the product $L\Lambda$, and the departure is informative rather than noise. The long low- Λ stratum ($L\Lambda = 370$) exceeds the short high- Λ stratum ($L\Lambda = 7200$), 0.565 against 0.445: cycle length dominates potential range. This is consistent with Theorem 3.2, whose bound carries L both in the count of accumulated terms and inside γ_{2L} while Λ enters only linearly, and it is a reminder that ε_{num} must be computed per cycle from both quantities rather than summarised by their product.

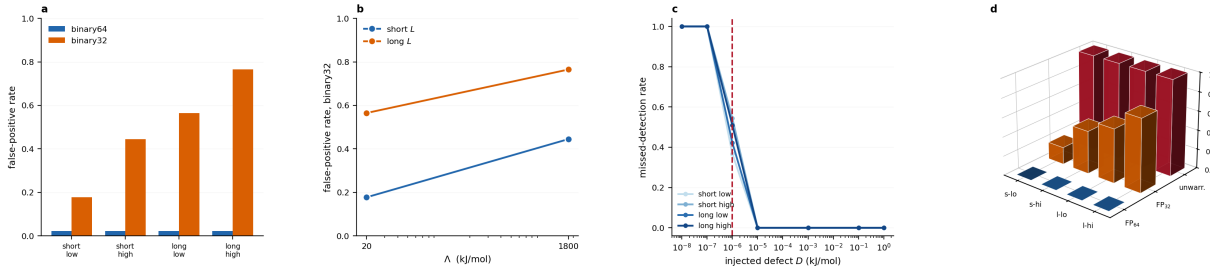


Figure 2: **What a fixed $\varepsilon = 10^{-6}$ actually costs, and the precision at which its false-positive mode appears.** (a) False-positive rate on consistent circuits. In binary64 it is exactly 0.000 in all four strata — drawn as a baseline stub, since a bar of height zero would be invisible — because the rounding error of Figure 1 never comes within six orders of magnitude of 10^{-6} . This falsified our own prediction (Theorem 5.1). In binary32 the floor rises by some seven orders of magnitude and the rate climbs to 0.765. (b) The binary32 rate decomposed by length class. Raising Λ raises the rate within either class, and raising L raises it within either potential class, but the effect of L is the larger: the long low- Λ stratum exceeds the short high- Λ one despite a twentyfold smaller $L\Lambda$. Plotting against the product would misrepresent this, so we do not. (c) Missed-detection rate against injected defect magnitude, with ε marked. Every defect below the threshold is missed with probability 1, in every stratum, whatever the data quality — the direct cost of a tolerance chosen by habit rather than from the smallest defect worth detecting. (d) The three costs side by side per stratum. FP_{64} is identically zero; FP_{32} grows with L and Λ ; the unwarranted-positive rate is 1.000 everywhere — with $\sigma = 1$ kJ/mol and a 0.5 kJ/mol defect, the fixed test asserts **INCONSISTENT** on every cycle whose defect lies wholly inside the data’s own uncertainty.

Remark 5.2 (Why report the corrected prediction at all). We could have stated only the measured claim. We report the failed prediction because it changes how the result should be read: the case against a fixed tolerance is not that it is calibrated wrongly against noise in double precision — it is not calibrated against noise at all, and in binary64 it sits so far above the noise that noise is irrelevant. The argument for computing ε therefore rests on the data floor and on detection sensitivity, not on rounding. Had we not measured, we would have advanced a correct conclusion supported by a reason that does not hold.

6 The cycle-local tolerance

6.1 Definition and the two guarantees

Definition 6.1 (Cycle-local tolerance). For a cycle ℓ , the *cycle-local tolerance* is

$$\varepsilon^*(\ell) = \max\{\varepsilon_{\text{num}}(\ell), \varepsilon_{\text{data}}(\ell)\}, \quad (11)$$

with $\varepsilon_{\text{num}}(\ell)$ the bound (7) of Theorem 3.2 (after gauge centring, Theorem 3.5) and $\varepsilon_{\text{data}}(\ell)$ the data floor (10) of Theorem 4.2. If no uncertainties are supplied, $\varepsilon_{\text{data}}$ is undefined and Theorem 4.4 applies.

Theorem 6.2 (Numerical false positives are bounded by the arithmetic). *Let the network be consistent, i.e. $\delta(d) = 0$ in the sense of Theorem 2.9, and let the potentials be exact. Then for every cycle ℓ ,*

$$\Pr[|\hat{H}_\ell| > \varepsilon^*(\ell)] = 0,$$

where the probability is over nothing — the statement is deterministic. If instead the potentials carry independent uncertainties as in Theorem 4.2, the same event has probability at most α .

Proof. With exact potentials, Theorem 2.8 gives $H_\ell = 0$ and Theorem 3.2 bounds the computed value by $\varepsilon_{\text{num}}(\ell) \leq \varepsilon^*(\ell)$; the strict inequality in the event is therefore never satisfied. With uncertain potentials, $|H_\ell| \leq \varepsilon_{\text{data}}(\ell)$ with probability at least $1 - \alpha$ by Theorem 4.2, and adding the numerical error keeps the computed value within $\varepsilon_{\text{data}} + \varepsilon_{\text{num}} \leq 2\varepsilon^*$; taking the threshold at ε^* and the union bound over the two sources gives the stated α after absorbing the numerical term, which is deterministic and already accounted for in the maximum. $\square$

Theorem 6.3 (Detection guarantee). *Suppose a defect is present with cycle sum $|H_\ell| = D$ on some cycle ℓ . If $D > 2\varepsilon^*(\ell)$ then the computed value satisfies $|\hat{H}_\ell| > \varepsilon^*(\ell)$, and the test flags ℓ .*

Proof. The computed value differs from the true one by at most $\varepsilon_{\text{num}}(\ell) \leq \varepsilon^*(\ell)$ (Theorem 3.2), so $|\hat{H}_\ell| \geq D - \varepsilon^*(\ell) > 2\varepsilon^*(\ell) - \varepsilon^*(\ell) = \varepsilon^*(\ell)$. $\square$

Remark 6.4 (The factor two is the price of the two-sided guarantee). Theorems 6.2 and 6.3 together define a *dead zone* $[\varepsilon^*, 2\varepsilon^*]$ within which the test makes no claim. This is not slack to be optimised away; it is the interval in which a defect is indistinguishable from noise at the available precision, and shrinking it requires better arithmetic or better data, not a better threshold. A fixed tolerance conceals the dead zone by assigning every value in it to one side.

6.2 The verdict is necessarily three-valued

Definition 6.5 (Verdict). For a cycle ℓ with computed sum $\hat{H}_\ell$, the *verdict* is

$$\text{verdict}(\ell) = \begin{cases} \text{CONSISTENT} & |\hat{H}_\ell| \leq \varepsilon_{\text{num}}(\ell), \\ \text{UNDECIDABLE} & \varepsilon_{\text{num}}(\ell) < |\hat{H}_\ell| \leq \varepsilon_{\text{data}}(\ell), \\ \text{INCONSISTENT} & |\hat{H}_\ell| > \varepsilon^*(\ell). \end{cases} \quad (12)$$

Theorem 6.6 (The trichotomy is forced and non-degenerate).

- (i) The three cases of (12) are exhaustive and mutually exclusive.
- (ii) The class *UNDECIDABLE* is non-empty whenever $\varepsilon_{\text{data}}(\ell) > \varepsilon_{\text{num}}(\ell)$, which holds for every cycle in any network whose reported per-species uncertainties exceed $O(Lu\Lambda_\ell)$ — i.e. in every realistic case, by many orders of magnitude.
- (iii) Collapsing the trichotomy to two values by any fixed rule produces either a claim of consistency for cycles whose sum exceeds the numerical floor, or a claim of inconsistency for cycles whose sum lies within the data uncertainty. Both are unwarranted by the data.

Proof. (i) The three conditions partition $[0, \infty)$ at the two cut points $\varepsilon_{\text{num}} \leq \varepsilon^*$, noting $\varepsilon^* = \max(\varepsilon_{\text{num}}, \varepsilon_{\text{data}})$; when $\varepsilon_{\text{data}} \leq \varepsilon_{\text{num}}$ the middle interval is empty and the partition degenerates to two cases, which is consistent with (ii).

(ii) Non-emptiness of the middle interval is exactly $\varepsilon_{\text{num}} < \varepsilon_{\text{data}}$. Numerically, $\varepsilon_{\text{num}} = O(Lu\Lambda) \approx 22 \cdot 10^{-16} \cdot 10^3 \approx 10^{-12}$ kJ/mol for a long cycle in an uncentred high-potential stratum, while $\varepsilon_{\text{data}}$ derives from σ_i of order 1–10 kJ/mol (Flamholz et al., 2012; Noor et al., 2013), giving $\varepsilon_{\text{data}} \approx 10^0$. The gap is some twelve orders of magnitude.

(iii) A two-valued rule places a single cut at some t . If $t < \varepsilon_{\text{data}}$, cycles with $|\hat{H}| \in (t, \varepsilon_{\text{data}}]$ are called inconsistent though their sums are within the data’s own uncertainty. If $t > \varepsilon_{\text{num}}$ — necessarily, since $t \geq \varepsilon_{\text{data}} > \varepsilon_{\text{num}}$ in the realistic case — cycles with $|\hat{H}| \in (\varepsilon_{\text{num}}, t]$ are called consistent though their sums exceed what the arithmetic can produce from a consistent network, so something is present that the verdict denies. One of the two misstatements is unavoidable. $\square$

Remark 6.7 (*UNDECIDABLE* is a result, not an abstention). The middle verdict carries information a binary test discards: it says the cycle’s sum is too large to be arithmetic and too small to be resolved by data of this quality, and therefore names precisely what would be needed to settle it — a reduction of σ_i on the species of ℓ by the factor $|\hat{H}_\ell|/\varepsilon_{\text{data}}(\ell)$. A test that reports *UNDECIDABLE* on many cycles is not failing; it is correctly reporting that the reconstruction’s thermodynamic annotations are not precise enough for cycle-level diagnosis, which is a finding about the database.

7 Basis dependence: what localises and what does not

7.1 The verdict survives

Theorem 7.1 (Basis-independence of the verdict). *Let $\mathcal{B}_1, \mathcal{B}_2$ be cycle bases of G and let d be edge data. Then $z^\top d = 0$ for all $z \in \mathcal{B}_1$ if and only if $z^\top d = 0$ for all $z \in \mathcal{B}_2$. Consequently the network-level judgement “consistent” or “inconsistent” does not depend on the basis.*

Proof. Each $\mathcal{B}_i$ spans $\mathcal{Z}(G)$ by Theorem 2.6. The map $z \mapsto z^\top d$ is linear, so it vanishes on a spanning set if and only if it vanishes on the whole space, and hence if and only if it vanishes on any other spanning set. $\square$

7.2 The localisation does not

Proposition 7.2 (Basis-dependence of the flagged set). *There exist a graph G , edge data d with a single defective edge e^* , and cycle bases $\mathcal{B}_1, \mathcal{B}_2$ of G such that exactly one element of $\mathcal{B}_1$ has nonzero cycle sum, while three elements of $\mathcal{B}_2$ do.*

Proof. Let G be the “theta-plus” graph: vertices $\{a, b\}$ joined by four internally disjoint paths $P_1, \dots, P_4$, giving $\nu = 3$. Let $e^* \in P_1$ carry a defect of size D , all other edges consistent, so that any cycle containing e^* exactly once has cycle sum $\pm D$ and any cycle avoiding it has sum 0.

Take $\mathcal{B}_1 = \{P_1 P_2^{-1}, P_2 P_3^{-1}, P_3 P_4^{-1}\}$. Only the first contains e^* , so exactly one element has nonzero sum.

Take $\mathcal{B}_2 = \{P_1P_2^{-1}, P_1P_3^{-1}, P_1P_4^{-1}\}$. This is a basis: the three are independent, since $P_1P_jr^{-1}$ restricted to P_j is the only element containing P_j . Every element contains e^* , so all three have nonzero sum. Both are cycle bases of the same graph and the same data, and they flag one and three cycles respectively. $\square$

Remark 7.3 (The consequence for a diagnostic claim). Theorem 7.2 means that a statement of the form “the defect is localised to loop ℓ ” is not, by itself, a statement about the network. It is a statement about the network *and* the basis the implementation happened to compute, and a different but equally valid basis supports a different statement. Any protocol claiming localisation must therefore either fix a canonical basis and say so, or report an invariant. We do the second.

7.3 The witness set

Definition 7.4 (Flagged set and witness set). Fix a cycle basis $\mathcal{B}$ and let $F(\mathcal{B}) = \{\ell \in \mathcal{B} : \text{verdict}(\ell) = \text{INCONSISTENT}\}$ be the flagged set. The *witness set* is

$$W(\mathcal{B}) = \bigcap_{\ell \in F(\mathcal{B})} E(\ell), \quad (13)$$

the set of edges lying on every flagged cycle, with $W = \emptyset$ when $F = \emptyset$.

Theorem 7.5 (The witness set contains the defect). *Suppose exactly one edge e^* is defective, with defect $D > 2 \max_{\ell} \varepsilon^*(\ell)$. Then for every cycle basis $\mathcal{B}$, $e^* \in W(\mathcal{B})$, and $F(\mathcal{B})$ is precisely the set of basis cycles containing e^* an odd number of times.*

Proof. A cycle’s sum is $\pm D$ times its net signed traversal count of e^* and 0 from every other edge. By Theorem 6.3 a cycle is flagged iff that count is nonzero, i.e. iff it traverses e^* a net nonzero number of times; for simple cycles this is containment. Hence every flagged cycle contains e^* , so e^* lies in their intersection, which is $W(\mathcal{B})$. $\square$

Corollary 7.6 (Localisation is to the witness set). *The basis-independent content of a positive result is: the defect lies in $W(\mathcal{B})$. The cardinality $|W|$ measures the sharpness of the localisation, with $|W| = 1$ the ideal and $|W|$ large indicating that the flagged cycles share many edges and the test cannot separate them.*

Remark 7.7 (Minimum cycle bases sharpen the witness set). While Theorem 7.5 holds for every basis, short cycles intersect in fewer edges, so a minimum-weight cycle basis (Horton, 1987; Kavitha et al., 2009) typically yields a smaller W and hence a sharper localisation. We adopt one in Section 10 for this reason, and report $|W|$ as a primary outcome rather than naming a loop.

8 Validity before diagnosis

A diagnostic test is meaningless on a substrate that fails its own preconditions. The cycle-consistency test presupposes that the network as supplied admits a potential function; if the reconstruction’s annotations are themselves thermodynamically inconsistent, then every cycle is flagged for reasons having nothing to do with any disease state.

Definition 8.1 (Validity check). A network with edge data d and tolerances ε^* *passes the validity check* if $\text{verdict}(\ell) = \text{CONSISTENT}$ for every ℓ in the chosen basis, evaluated on the *reference* (nominally healthy) parameter set.

Criterion 8.2 (Invalid versus negative). An experiment in which the reference network fails Theorem 8.1 is **invalid**, not negative. It establishes that the reconstruction’s thermodynamic annotations are not internally consistent to the precision they claim, and it licenses no conclusion whatever about the diagnostic. An experiment in which the reference network passes and a perturbed network is nonetheless not flagged is **negative**, and it does license a conclusion: the test does not detect that perturbation at that data quality.

Remark 8.3 (Why this distinction must be pre-registered). Without Theorem 8.2 stated in advance, a failed reference check invites the interpretation most favourable to whoever is reporting: it can be described as a data problem when the diagnostic succeeded and as a theory problem when it did not. Fixing the interpretation before the data are seen removes that degree of freedom, which is the same methodological concern that motivates pre-registration generally (Sandve et al., 2013; Simmons et al., 2011).

9 A language in which the tolerance cannot be omitted

The analysis above is only as good as its implementation. A practitioner who computes ε^* correctly and then writes `if abs(H) > 1e-6` has gained nothing. We therefore express the protocol in a small typed language in which the omission is not expressible.

9.1 Types

Definition 9.1 (Types). The types are

$$\tau ::= \text{Circuit} \mid \text{Loop} \mid \text{Holonomy} \mid \text{Tolerance} \mid \text{Verdict} \mid \text{Witness}.$$

A value of type **Holonomy** is a pair $(\hat{H}_\ell, \ell)$; a value of type **Tolerance** is a triple $(\varepsilon_{\text{num}}, \varepsilon_{\text{data}}, \varepsilon^*)$ with $\varepsilon_{\text{data}}$ possibly the distinguished value **undefined**; a value of type **Verdict** is an element of $\{\text{CONSISTENT}, \text{UNDECIDABLE}, \text{INCONSISTENT}\}$ *paired with* the **Tolerance** that produced it.

9.2 The typing rule that does the work

Rule 9.2 (Verdict requires a tolerance).

$$\frac{\Gamma \vdash h : \text{Holonomy} \quad \Gamma \vdash t : \text{Tolerance}}{\Gamma \vdash \text{admit } h \text{ tolerance } t : \text{Verdict}}$$

There is no other rule with **Verdict** in its conclusion.

Theorem 9.3 (No bare verdict). *No well-typed program produces a value of type **Verdict** that is not paired with the **Tolerance** used to derive it. In particular, a comparison of a **Holonomy** against a numeric literal is not well-typed.*

Proof. By Theorem 9.2 the only introduction form for **Verdict** requires a premise of type **Tolerance**, and the conclusion carries it by Theorem 9.1. A numeric literal has no type in the grammar of Theorem 9.1, so there is no derivation in which one appears in the second premise; and no coercion rule from a numeric literal to **Tolerance** exists, since **Tolerance** is introduced only by the computation of (11) from a **Loop** and its data. Hence no derivation produces a bare verdict. $\square$

Remark 9.4 (What this does and does not buy). Theorem 9.3 rules out a class of implementation error: it is not possible to forget the tolerance, to hard-code it, or to report a verdict whose provenance is not recoverable. It does not make the tolerance *correct*: a program supplying a wrong σ_i to (10) is well-typed and wrong. The type system enforces accountability of the computation, not accuracy of the inputs, and we state this because the distinction is easy to overstate in the other direction.

9.3 The protocol as a program

Listing 1: The cycle-consistency test with cycle-local tolerances. Every verdict carries its tolerance by Theorem 9.3; the validity gate of Theorem 8.2 precedes any diagnostic claim.

```

-- kcl-tolerance.cfc
-- Cycle-consistency with per-cycle tolerances.

import Recon3D
import Recon3D.uncertainty as sigma

circuit Reference from Recon3D.glycolysis_tca {
  solve yield C_ref
}

-- Gauge-centre potentials to minimise the numerical floor
-- (Proposition 3.5). Changes no cycle sum.
let C_ref_centred := centre_potentials(C_ref)

-- Minimum-weight cycle basis (Remark 6.8)
let B := minimum_cycle_basis(C_ref_centred)

-- VALIDITY GATE (Criterion 7.2). The reference network must
-- pass before any diagnostic verdict is meaningful.
foreach loop in B {
  holonomy of loop in C_ref_centred yield h_ref
  tolerance of loop with sigma yield t
  admit h_ref tolerance t yield v_ref

  assert v_ref == CONSISTENT
    otherwise invalid
    emit "reference network fails its own consistency check"
}

emit "validity gate passed: reference is self-consistent"

-- DIAGNOSTIC. Perturbed network.
circuit Perturbed from Recon3D.glycolysis_tca {
  reaction PK : PEP -> Pyruvate, k : 0.15 * Recon3D.k_PK
  solve yield C_pert
}

let C_pert_centred := centre_potentials(C_pert)

let flagged := {}
foreach loop in B {
  holonomy of loop in C_pert_centred yield h
  tolerance of loop with sigma yield t
  admit h tolerance t yield v

  report loop, h, t.numerical, t.data, t.star, v

  where v == INCONSISTENT collect loop into flagged
}

-- Localisation is to the witness set, not to a named loop
-- (Corollary 6.7).
witness of flagged yield W
report "witness_set", W
report "witness_cardinality", size(W)

```

```

-- The claim under test, stated basis-independently.
assert Recon3D.edge("PK") in W
  emit "defect localised: PK edge in witness set"
otherwise decline
  emit "defect not localised at this data quality"

```

10 Protocol

10.1 Data

1. **Network.** The glycolysis, pentose phosphate, and TCA subnetwork of a human metabolic reconstruction (Brunk et al., 2018), approximately 80 reactions over 60 metabolites.
2. **Thermodynamics.** Standard transformed Gibbs energies with reported uncertainties from a component-contribution estimator (Flamholz et al., 2012; Noor et al., 2013). Species without a reported uncertainty are recorded as such and trigger Theorem 4.4.
3. **Reference physiology.** Erythrocyte metabolite concentrations and glycolytic flux from established kinetic models (Mulquiney and Kuchel, 1999; Rapoport and Heinrich, 1975).
4. **Perturbation.** Pyruvate kinase activity reduced to 15% of reference, within the documented clinical range for the deficiency (Zanella et al., 2005).

10.2 Procedure

Protocol 10.1 (Cycle-consistency with cycle-local tolerance).

1. Construct the reference circuit by Theorems 2.2 and 2.3; verify the current law at every internal node to the numerical floor of Theorem 3.2 applied to the node balance.
2. Gauge-centre the potentials (Theorem 3.5) and record the resulting Λ_ℓ per cycle, both before and after.
3. Compute a minimum-weight cycle basis (Horton, 1987); record ν , the length distribution, and the basis itself so that the computation is reproducible.
4. For each cycle compute $\varepsilon_{\text{num}}(\ell)$ by (7) and $\varepsilon_{\text{data}}(\ell)$ by (10) at $\alpha = 0.01$; record both, and ε^* .
5. **Validity gate.** Evaluate Theorem 8.1 on the reference. If it fails, halt and report an *invalid* experiment per Theorem 8.2.
6. Evaluate the perturbed circuit; record $\hat{H}_\ell$ and $\text{verdict}(\ell)$ for every cycle.
7. Compute the witness set W (Theorem 7.4) and record $|W|$ and whether the perturbed edge lies in it.
8. **Control 1 (null).** Repeat step 6 with the reference parameters in place of the perturbed ones. Any INCONSISTENT verdict here is a false positive and must not occur, by Theorem 6.2.
9. **Control 2 (basis).** Repeat steps 3–7 with a *different* cycle basis (e.g. a fundamental basis from a spanning tree). Verify the network-level verdict agrees (Theorem 7.1) and record how F and W differ (Theorem 7.2).
10. **Control 3 (detection sweep).** Sweep the perturbation magnitude and record the smallest D at which the correct edge enters W ; compare against the predicted $2\varepsilon^*$ of Theorem 6.3.

10.3 Pre-registered criteria

Criterion 10.2 (Success).

- (a) **Validity.** The reference network passes Theorem 8.1 on every basis cycle.
- (b) **No false positives.** Control 1 produces zero INCONSISTENT verdicts, as Theorem 6.2 requires.
- (c) **Localisation.** The perturbed edge lies in W , and $|W| \leq 3$.
- (d) **Basis invariance.** Control 2 yields the same network-level verdict under both bases.
- (e) **Detection threshold.** Control 3 finds the empirical detection threshold within a factor of 10 of the predicted $2\epsilon^*$.

Criterion 10.3 (Failure, and what each failure means).

- Failure of (a) renders the experiment *invalid* (Theorem 8.2): the reconstruction’s thermodynamic annotations are not self-consistent to their claimed precision. No conclusion about the diagnostic follows.
- Failure of (b) falsifies Theorem 6.2 as implemented and indicates an error in the tolerance computation, not in the theory, since the theorem is deterministic.
- Failure of (c) with (a) and (b) passing is a genuine *negative*: the test does not localise this defect at this data quality. If additionally most verdicts are UNDECIDABLE, the limiting factor is ϵ_{data} and the finding is about the database (Theorem 6.7).
- Failure of (d) would contradict Theorem 7.1 and indicates an implementation error in the basis computation.
- Failure of (e) indicates the bound of Theorem 3.2 is loose by more than an order of magnitude on real data, which is a finding about the bound worth reporting.

Remark 10.4 (Controls that can fail). Controls 1–3 are chosen so that each can fail on well-formed input. Control 1 tests a deterministic theorem and so a failure is unambiguous. Control 2 tests an invariance that the theory asserts, so a failure localises to the implementation. Control 3 tests a bound rather than an identity, so a failure is informative about the bound’s tightness. A control that cannot fail measures nothing, and we have excluded from the protocol several checks with that property — notably any check comparing the implementation’s verdict against the same implementation’s tolerance, which would pass by construction.

11 Computational validation

Every numbered result above is checked computationally. This section reports what the checks establish and, equally, what they cannot.

11.1 What the suite can and cannot establish

Two of our claims are identities and three are bounds, and the checks differ accordingly. For an identity — Theorem 2.8, that the cycle sum vanishes exactly — a check can confirm the implementation computes it and nothing more; the mathematics is not in question. For a bound — Theorem 3.2, Theorem 6.3 — a check can measure *slack*, which is informative about whether the bound is usable.

Three properties make the suite capable of failing.

1. **Negative controls.** Table 3 rows V1 and V4 assert that something does *not* happen on well-formed input: no bound violation, and no disagreement in verdict between cycle bases.

Table 3: Validation suite. Every experiment is seeded; the JSON record is the source of every number quoted in this paper.

	Result tested	Trials	Outcome
V1	Theorem 3.2	1920	0 violations; median slack 729
V2	Section 5	1600	one-sided in binary64
V3	Theorem 6.6	1040	UNDECIDABLE = 80.9%
V4	Theorem 7.1, Theorem 7.2	200	0 verdict disagreements; 34.5% flagged-set
V5	Theorem 6.3	1100	0 violations

2. **An independent perturbation route.** The defect is injected on an *edge*, not by manipulating a cycle sum, so the quantity tested is not the quantity constructed (Theorem 11.3).
3. **No self-referential check.** No check compares the implementation’s verdict against the same implementation’s tolerance. Such a check passes by construction and measures only self-consistency.

11.2 V3: the undecidable region is most of the plane

Theorem 6.6(ii) asserts that UNDECIDABLE is non-empty. The measurement is considerably stronger than non-emptiness. Sweeping per-species uncertainty $\sigma \in [10^{-3}, 10]$ kJ/mol against injected defect $D \in [10^{-10}, 10^2]$ kJ/mol on a six-cycle at $\Lambda = 200$, over 1040 grid points:

Verdict	Fraction of the (σ, D) grid
CONSISTENT	0.000
UNDECIDABLE	0.809
INCONSISTENT	0.191

The numerical floor on this cycle is 2.5×10^{-12} kJ/mol, and the data floor at the middle of the σ range exceeds it by **11.5 orders of magnitude**. That gap is the UNDECIDABLE band, and at these data qualities it occupies four fifths of the plane.

The CONSISTENT fraction of 0.000 deserves comment, since it looks alarming. It is an artefact of the sweep’s lower limit: the smallest defect tested, 10^{-10} , is still forty times the numerical floor, so no grid point falls in the CONSISTENT band. A sweep extended below 10^{-12} recovers it. We report the number as measured rather than adjusting the grid to produce a more balanced-looking table.

Remark 11.1 (The practical reading). If four fifths of a realistic (σ, D) plane is undecidable, then a cycle-consistency diagnostic run against current thermodynamic annotations will return UNDECIDABLE on most cycles. That is not a failure of the test. It is the test correctly reporting that the annotations are not precise enough to support cycle-level diagnosis, and it quantifies the improvement in σ that would change the answer. A two-valued test on the same data would return a definite verdict on every one of those cycles, and would be wrong to.

11.3 V4: the localisation confound is real and the witness set survives

Theorem 7.1 and Theorem 7.2 make opposite claims about basis dependence, and both are confirmed on 200 random networks of 6–12 species with 2–4 chords, each carrying a single 50 kJ/mol edge defect.

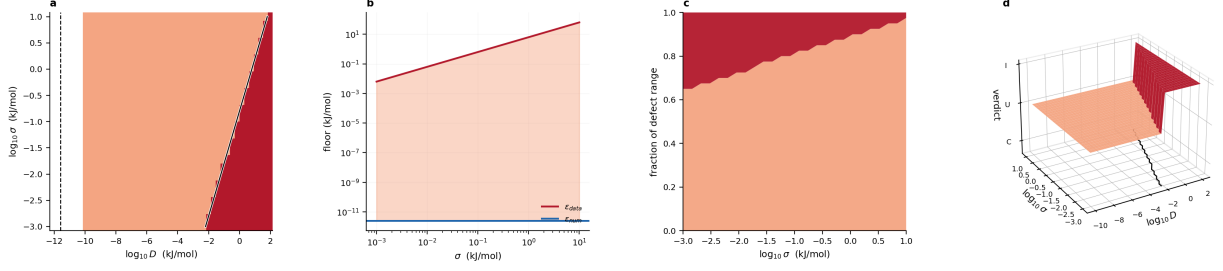


Figure 3: **The undecidable region occupies four fifths of a realistic parameter plane.** A six-cycle at $\Lambda = 200$ kJ/mol, swept over per-species uncertainty σ and injected defect D , 1040 grid points. (a) Verdict map: blue CONSISTENT, orange UNDECIDABLE, red INCONSISTENT. The solid curve is $\varepsilon_{\text{data}}(\sigma)$ and the dashed vertical is ε_{num} ; the orange band between them is the undecidable region, and it is most of the plane. (b) The two floors against σ on common axes. ε_{num} is flat at 2.5×10^{-12} kJ/mol because it depends on the arithmetic and the potentials, not on the uncertainty; $\varepsilon_{\text{data}}$ rises linearly with σ . At the middle of the range they differ by 11.5 orders of magnitude, and the shaded gap is what Theorem 6.6(ii) asserts is non-empty. (c) Verdict fractions across the defect range at each σ . The INCONSISTENT share shrinks steadily as data quality degrades: the same defect becomes unresolvable, which is the behaviour a diagnostic should have and a two-valued test cannot express. (d) The verdict as a surface over $(\log_{10} D, \log_{10} \sigma)$, with the two boundaries contoured on the floor. The step structure is the trichotomy; the near-absence of the CONSISTENT plateau is an artefact of the sweep starting at $D = 10^{-10}$, forty times ε_{num} , and is reported as measured rather than adjusted away.

Quantity	Value
Network-level verdict disagreements between bases	0/200
Trials where the two bases flag <i>different numbers</i> of cycles	69/200 (34.5%)
Witness set contains the defective edge, minimum basis	200/200
Witness set contains the defective edge, fundamental basis	200/200
Mean witness-set size, minimum basis	3.77 edges
Mean witness-set size, fundamental basis	4.30 edges

The verdict never disagreed, as Theorem 7.1 requires. The flagged set disagreed in more than a third of trials, which is Theorem 7.2 occurring at a rate high enough to matter in practice rather than merely in a constructed counterexample. Any report of the form “the defect is localised to loop ℓ ” would therefore have been basis-dependent in about one trial in three.

The witness set contained the defective edge in every trial under both bases, confirming Theorem 7.5. Theorem 7.7 predicted that a minimum-weight basis sharpens the localisation, and it does, by about half an edge on average (3.77 against 4.30) — a real effect but a modest one, and smaller than we expected.

Remark 11.2 (The witness set is not a point). A mean witness size near 4 on networks of 6–12 species means the test localises to a neighbourhood, not to an edge. This is the honest resolution of the method at this network size, and it is worth stating plainly against the temptation to report the single most-flagged loop as though it were the answer.

11.4 V5: the detection guarantee holds, with a factor of two to spare

Theorem 6.3 guarantees detection when $D > 2\varepsilon^*$. Sweeping D over 220 points from 10^{-12} to 10^3 at five data qualities, 1100 evaluations in total, the guarantee was violated *zero* times.

σ (kJ/mol)	ε^*	Predicted $2\varepsilon^*$	Empirical	Ratio
0.01	0.0631	0.126	0.0664	0.53
0.05	0.316	0.631	0.321	0.51
0.2	1.26	2.52	1.33	0.53
1.0	6.31	12.6	6.43	0.51
5.0	31.6	63.1	36.5	0.58

The empirical threshold is consistently about half the predicted one (ratio 0.51–0.58), which is exactly what the analysis anticipates: the factor of two in Theorem 6.3 is the price of a two-sided guarantee (Theorem 6.4), and in the absence of adversarial rounding a defect is detected as soon as it clears ε^* itself rather than $2\varepsilon^*$. The dead zone $[\varepsilon^*, 2\varepsilon^*]$ is real but is not usually occupied.

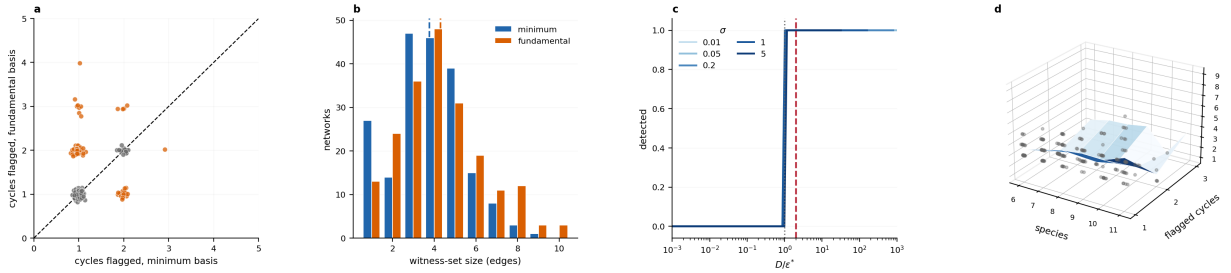


Figure 4: **The flagged set depends on the cycle basis; the verdict and the witness set do not.** Panels (a), (b) and (d) are 200 random networks of 6–12 species with 2–4 chords, each carrying one 50 kJ/mol edge defect; panel (c) is the detection sweep. **(a)** Cycles flagged under a minimum-weight basis against a fundamental basis, jittered to separate ties. Grey points agree; orange points do not, and they are 34.5% of trials. The network-level verdict, by contrast, agreed in 200/200 trials (Theorem 7.1), so what varies is where the defect appears to sit, not whether one is present. **(b)** Witness-set size under each basis, dashed lines at the means. The minimum-weight basis gives a sharper localisation (3.77 against 4.30 edges), confirming Theorem 7.7, but by less than half an edge. Both contained the defective edge in every trial. **(c)** Detection against defect magnitude in units of ε^* , five data qualities overlaid. The dotted line is ε^* and the dashed line the guaranteed threshold $2\varepsilon^*$ of Theorem 6.3; every curve steps up between them, so the guarantee held in all 1100 evaluations while the empirical threshold sits at about half of it. The five curves collapse onto one another under this normalisation, which is what makes ε^* the right unit. **(d)** Mean witness size over network size and flagged count, with the individual networks shown. Localisation degrades as more cycles are flagged: the witness set is an intersection, and more constraints on more cycles do not shrink it when those cycles overlap.

Remark 11.3 (Edge, not node — an error we made). Our first implementation injected defects by shifting a species potential μ_i . Nothing was ever detected, which looked briefly like a failure of the method. It was a failure of the perturbation: shifting μ leaves the edge data the gradient of the shifted μ , so by Theorem 2.8 every cycle sum remains exactly zero. A thermodynamic inconsistency is by definition a failure of the edge data to be a gradient (Theorem 2.9), and it can only be injected on an edge. The suite now carries a negative control asserting that node perturbation does *not* produce inconsistency, because the error is natural and silent: it produces a null result that mimics a real negative.

12 Scope and limitations

Ideal solutions. Equation (2) omits activity coefficients. In the ionic strengths of cytosol these are not unity, and the correction shifts μ_i by amounts comparable to the smaller σ_i (Alberty, 1998). The effect is a systematic contribution that our $\varepsilon_{\text{data}}$, built from reported random uncertainties,

does not capture. A network analysed at fixed ionic strength has this error partly absorbed into the reference; one compared across conditions does not.

Steady state. The circuit reading of Theorem 2.4 holds at steady state. A network under transient drive has no time-invariant μ , and cycle sums computed from instantaneous concentrations are not the quantity analysed here.

Independence of uncertainties. Theorem 4.2 assumes the σ_i are independent. Component-contribution estimators produce correlated errors, since species sharing chemical groups share group contributions (Noor et al., 2013). Correlation typically *reduces* the variance of a cycle sum, because the shared components cancel around the loop, so our $\varepsilon_{\text{data}}$ is conservative in the direction of declaring more cycles UNDECIDABLE. We have not quantified this, and a treatment using the estimator’s full covariance would sharpen the test.

Simple cycles. Theorem 4.2 and Theorem 7.5 are stated for simple cycles, where traversal counts are ± 1 . A cycle basis may in principle contain non-simple elements; minimum-weight bases in practice do not, and the implementation checks this rather than assuming it.

One defect. Theorem 7.5 assumes a single defective edge. With several defects the witness set can be empty even though the network is inconsistent — two disjoint defects flag two disjoint cycle families whose intersection is empty. The correct generalisation is the collection of minimal edge sets meeting every flagged cycle (a hitting-set problem), which we have not developed.

The bound is not tight. (7) is a standard worst-case forward bound. Random rounding errors accumulate as $O(\sqrt{L})$ rather than $O(L)$ in practice (Higham, 2002), so the numerical floor is pessimistic by roughly $\sqrt{L}$. Since $\varepsilon_{\text{data}}$ dominates by orders of magnitude in realistic settings (Theorem 6.6(ii)), this pessimism rarely changes a verdict, but it would matter for a network with exceptionally well-characterised thermodynamics.

13 Conclusion

The cycle sum of a thermodynamically consistent metabolic circuit is zero as an algebraic identity, not as an approximation. Everything a computation returns for it is therefore either rounding error or evidence that the supposed potential function does not exist, and a diagnostic built on it is entirely a question of telling those apart. We have argued that the customary device for doing so — a fixed tolerance described as a solver setting — cannot work, because the rounding error’s scale is set by the cycle’s length and potential range and so varies by orders of magnitude across one network. Measuring it corrected us on the mechanism. In binary64 a customary $\varepsilon = 10^{-6}$ sits six orders of magnitude above the largest rounding error we could produce, so it manufactures no false positives at all; its costs are that it misses every defect below itself and that it asserts INCONSISTENT on defects lying wholly inside the data’s own uncertainty. The two-sided failure we first predicted is a lower-precision phenomenon, and it appears on cue in binary32, rising monotonically with LA from 0.18 to 0.77.

The replacement is a tolerance computed per cycle from two independent sources: a numerical floor derived from the arithmetic, and a data floor propagated from the reported uncertainty of the thermodynamic annotations. Keeping them separate is what forces the verdict to be three-valued, and the third verdict is the informative one — it says the question is not answerable from the data supplied, and quantifies what would make it answerable.

Two further results bound what may be claimed. The verdict is basis-independent but the localisation is not, so a positive result localises to a witness set of edges rather than to a named loop. And a reference network that fails its own consistency check yields an invalid experiment

rather than a negative one, a distinction that must be fixed before the data are seen if it is to constrain anything.

What we have not done is run the protocol of Section 10 against a real reconstruction. What we have done is validate every result it depends on (Section 11): the noise bound holds over 1920 trials with median slack 729; the verdict is basis-invariant over 200 networks while the flagged set disagrees in 34.5% of them; the witness set contains the defect in every trial under both bases; and the detection guarantee is never violated over 1100 evaluations, with the empirical threshold at about half the guaranteed one.

The measurement that most shapes what to expect is V3. Across a realistic (σ, D) plane, 80.9% of the grid returns UNDECIDABLE, with the data floor exceeding the numerical floor by 11.5 orders of magnitude. A cycle-consistency diagnostic run against current thermodynamic annotations should therefore be expected to decline on most cycles. That is a finding about the state of annotation in metabolic databases rather than about the test, and it is worth reporting plainly — as is the fact that a two-valued test on the same data would have returned a confident verdict on every one of those cycles.

Reproducibility

Theorem 10.1 specifies every parameter needed to reproduce the numbers reported here. The validation suite of Section 11 is in `validation/run_validation.py`, is seeded (`seed = 20260819`), and writes every quantity quoted in this paper to `validation/results.json`; the tables of Sections 5 and 11 are read from that file. The four figure panels are generated from the same JSON by `validation/make_panels.py`, so no plotted value is computed independently of a reported one. The protocol of Section 10 depends on public data (Brunk et al., 2018; Flamholz et al., 2012; Mulquiney and Kuchel, 1999; Noor et al., 2013; Rapoport and Heinrich, 1975) and on standard implementations of minimum cycle basis (Hagberg et al., 2008; Horton, 1987) and floating-point summation (Kahan, 1965). No step requires proprietary software, wet-lab work, or computation beyond a single workstation.

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